
WHEN IS A LEARNED COMMAND ADAPTER WORTH IT? CLOSED-LOOP IDENTIFICATION AND COUNTERFACTUAL AUDITING OF FROZEN LOCOMOTION POLICIES

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ABSTRACT

When a frozen, command-conditioned locomotion policy is deployed as a building block, the natural next step is a learned command adapter. We argue that this step should be preceded by a cheaper question: does the interface expose an improvement that is both real at the same state and recoverable from deployment-time features beyond a non-adaptive mixture of the same actions? We formalize an *adapter necessity audit* that separates three often conflated quantities—global operating-point gain, same-state counterfactual headroom, and state-allocation gain over a frequency-matched randomized mixture—and maps source-seed uncertainty into a GO/NO-GO/ABSTAIN decision under user-specified practical-value and violation thresholds. Closed-loop command-response identification supplies optional decision features: from excitation rollouts, a supervised model predicts situational motion, mechanical cost, and terrain-relative posture (displacement $R^2 \geq 0.92$, posture $R^2 \approx 0.9$, cost $R^2 \approx 0.75$ – 0.8). On Go2 and H1 the diagnosis is sharp. An episode-level oracle over three scales beats the best global scale by only 1.0–1.9%. Exact same-state replay on ten seeds finds 5.2% local work headroom under success and time constraints, yet a leave-one-seed-out treatment-effect selector recovers only 0.55% beyond its matched mixture; its candidate treatment-effect error (11.6%) exceeds the signal it must rank. This recoverability gap, not the absence of local opportunity, explains why neither a viability-gated compensator nor a heavier sampling MPC clears a tuned fixed-scaling frontier. Repeating Stages 2–3 on H1 (four seeds, observation-only) yields 4.2% oracle headroom but only 0.33% allocation gain and a $\delta = 1\%$ NO-GO. At a 1% practical-value threshold the ten-seed Go2 audit returns ABSTAIN (the allocation CI just crosses 1%), revising a five-seed NO-GO. Semi-synthetic controls on the same Go2 states calibrate the joint gain–constraint rule. The contribution is the audit: a counterfactual test of whether a learned adapter is supported by available signal, not another adapter claimed to clear that test.

Keywords adapter necessity audit · closed-loop system identification · frozen policies · counterfactual evaluation · legged locomotion

1 Introduction

Strong learned locomotion policies are increasingly deployed as frozen, command-conditioned building blocks: a task layer drives a pretrained velocity-tracking policy toward a goal through its command interface, typically with a hand-designed proportional command. In this regime the policy’s weights, reward, and training pipeline are unavailable or too costly to touch; one can only observe the policy acting and choose the next command. The usual response is to build a learned adapter. We ask a cheaper, logically prior question: does the interface expose an improvement that is real at the same state, and recoverable from deployment-time features beyond merely changing how often each command is issued?

Controller rankings alone cannot answer that question. Beating a proportional baseline may reflect a better global operating point rather than useful state-conditioned allocation; failing to beat a tuned fixed scale may mean either that

no local headroom exists or that the available representation cannot recover it. We therefore introduce an *adapter necessity audit* that separates three quantities: (i) global operating-point gain from choosing one fixed intervention; (ii) same-state counterfactual headroom of an oracle that observes realized outcomes; and (iii) state-allocation gain of a cross-fitted selector over a randomized mixture with identical action frequencies. Source-seed cluster bootstrap and user-specified practical-value / violation thresholds map these estimates into GO, NO-GO, or ABSTAIN. Abstention is first-class: an underpowered audit is not evidence that an adapter is unnecessary.

To supply optional decision features—and as a reusable measurement of the closed loop—we identify the frozen policy–robot response from excitation rollouts. A supervised model predicts, for any candidate command in the current situation, short-horizon motion, mechanical cost, and terrain-relative posture. Modeling the closed loop at the three-dimensional command interface, rather than action-level dynamics, makes identification a cheap supervised problem on the policy’s own rollouts. Average prediction accuracy is high (displacement $R^2 \geq 0.92$; posture $R^2 \approx 0.9$; cost $R^2 \approx 0.75$ – 0.8), and terrain-relative posture labels are necessary on rough ground. Accuracy alone, however, does not certify ranking value: the audit tests whether these predictions improve counterfactual selection beyond the observation itself.

As a concrete adapter under that audit we study **Viability-Gated Command Compensation (VGCC)**: a training-free controller that scores a structured candidate set around the proportional command and applies a bounded correction toward the predicted-cheapest eligible candidate, falling back exactly otherwise. Across three embodiments it lowers running cost and mechanical work relative to the proportional baseline, yet neither VGCC nor a heavier sampling MPC clears a tuned fixed-scaling frontier. The audit explains why. An episode-level oracle over three scales improves on the best global scale by only 1.0–1.9%. Exact same-state replay on ten source seeds finds 5.2% local work headroom under success and a 10% time budget, so local opportunity exists. Recoverability fails: a treatment-effect ensemble realizes only 0.55% beyond a frequency-matched mixture, while its candidate treatment-effect error (11.6%) exceeds the signal it must rank. At a 1% practical-value threshold the ten-seed audit returns ABSTAIN, revising a five-seed NO-GO.

Contributions. (1) *Adapter necessity audit.* A protocol that separates global operating-point gain, same-state counterfactual headroom, and recoverable state-allocation gain over a matched randomized mixture, with an explicit three-way deployment decision under practical-value and violation thresholds. (2) *Closed-loop command-response identification.* A data-efficient model of situational motion, mechanical cost, and terrain-relative posture for a frozen command-conditioned policy, with terrain-relative posture labels shown to be necessary and with identification treated as an optional feature source rather than an assumed prerequisite for the audit. (3) *Empirical diagnosis.* On standard rough-terrain reaching with two adapter designs and three embodiments, local headroom (5.2% Go2; 4.2% H1) is largely unrecoverable (0.55% / 0.33% allocation gain); Go2’s ten-seed audit abstains at $\delta = 1\%$ while H1’s four-seed observation-only audit returns NO-GO; semi-synthetic controls calibrate the joint gain–constraint rule. VGCC is the case study that motivates the audit, not a claimed frontier-dominating controller.

2 Related Work

Improving pretrained controllers without retraining. Residual policy learning adds action-space corrections but requires reward access and further interaction [Silver et al., 2018, Johannink et al., 2019]. RMA [Kumar et al., 2021] and multiplicity-of-behavior training [Margolis and Agrawal, 2022] expose adaptation at training time. Massively parallel pipelines and terrain curricula improve the base policy during training [Rudin et al., 2022, Makovychuk et al., 2021, Mittal et al., 2023, Lee et al., 2020, Miki et al., 2022]. Our setting is the remaining cell: the policy is frozen, only its command interface is writable, and the question is whether a learned adapter is justified. Interface-level optimization over a fixed low-level controller is successful in eco-driving [Sciarretta et al., 2015] and reference governors [Garone et al., 2017]; our audit asks when the analogous claim holds for efficiency over a velocity-commanded locomotion policy.

Closed-loop models for command-level control. Model-based RL typically learns action-level dynamics [Deisenroth and Rasmussen, 2011, Chua et al., 2018, Janner et al., 2019, Hafner et al., 2020, Hansen et al., 2022, Nagabandi et al., 2018]. When a competent policy already exists, modeling the closed loop at the command interface collapses input dimensionality and restricts decisions to commands the policy understands. Related instantiations include reference governors and predictive safety filters on prestabilized loops [Garone et al., 2017, Wabersich and Zeilinger, 2021], ABS searching twist commands against a co-trained reach-avoid value [He et al., 2024], and Koopman identification of an RL-controlled biped for safe navigation [Kim et al., 2024]. Feedforward/internal-model control [Åström and Murray, 2008, Francis and Wonham, 1976] and sampling MPC [Mayne et al., 2000, Williams et al., 2018, Kabzan et al., 2019, Di Carlo et al., 2018] supply the control templates. VGCC is a small member of this family—gated one-step compensation from an identified response model—used here as a case study. Unlike prior systems that treat

identification as a means to control, we treat it as a falsifiable feature source: we ask whether its predictions improve counterfactual ranking beyond the observation and beyond a frequency-matched non-adaptive mixture.

Treatment choice and policy value. Learning a command adapter from same-state interventions is a treatment assignment problem: context is observed before choosing an intervention, and the goal is policy value rather than marginal prediction accuracy. Empirical welfare maximization [Kitagawa and Tetenov, 2018], efficient policy learning [Athey and Wager, 2021], and counterfactual risk minimization [Swaminathan and Joachims, 2015] formalize this distinction. Our setting has deterministic exact-replay potential outcomes rather than unknown propensities, but adds robotics-specific requirements: task success/time eligibility and comparison against a randomized policy with the same intervention frequencies. The latter isolates state-conditioned allocation from globally changing the operating point—the distinction our audit is built to enforce.

Options and behavior archives. Temporal abstraction builds reusable behavior libraries [Sutton et al., 1999, Bacon et al., 2017, Eysenbach et al., 2019, Sharma et al., 2020, Peng et al., 2022, Mouret and Clune, 2015]. DADS plans over skill transition models co-trained with the skills [Sharma et al., 2020]; we instead identify an already-trained task policy. A retrieval-based archive variant is characterized in Appendix B.2 as coverage-limited relative to the parametric response model. We intervene at the command interface because raw chunk replay destabilizes contact-rich locomotion [Zhao et al., 2023].

3 Method

The method has three parts. The primary object is an *adapter value audit* that separates available counterfactual headroom from the value a state-conditioned selector recovers beyond a frequency-matched randomized policy (Section 3.3). Closed-loop command-response identification (Section 3.2) supplies optional decision features for that audit and for any compensator built on the same interface. VGCC (Section 3.4) is one concrete adapter placed under the audit: a bounded, gate-certified correction toward the cheapest of a structured candidate set. Identification is not assumed necessary for a positive audit decision; the audit measures whether its features help.

Design requirements. *R1, observation only:* weights, reward, and training pipeline are unavailable; everything must come from observing the policy act. *R2, non-degradation:* any deployed adapter must revert to the pretrained policy exactly when its intervention is uncertain. *R3, embodiment-light:* nothing may depend on robot-specific knowledge beyond what identification recovers from rollouts (posture thresholds may adapt to morphology; objectives and model architecture should not). These constraints shape the interface (Section 3.1), the response model, the audit protocol, and the VGCC case study.

3.1 Compensation Interface

Let π_θ be a frozen, pretrained continuous-control policy mapping observations o_t and a command c to low-level actions. In our locomotion instantiation $c = (v_x, v_y, \omega_z)$ is the body-frame velocity command, but the framework only requires *some* command interface through which behavior can be steered, a property shared by learned controllers deployed as task-level building blocks [Margolis and Agrawal, 2022, Kumar et al., 2021].

Scope. Three properties delimit where the framework applies. (i) *Command interface:* an unconditioned single-task policy exposes nothing to invert over. (ii) *Command redundancy:* several commands must make comparable progress at different cost; a fully determined command admits no efficiency gain. (iii) *Fast closed-loop response:* command consequences must be visible within a short prediction window (locomotion, end-effector servoing, vehicle tracking), not sparse long-horizon credit assignment. Section 5 returns to what lies outside these bounds.

We intervene at the command rather than the action for three reasons aligned with R1–R3. Action residuals require reward and further interaction; a command correction can be computed from an identified model. Command corrections leave π_θ in control of contacts, but still need explicit fallback and posture gating (R2). The command interface is shared across embodiments (R3).

A task supplies a target; throughout the method, g denotes that target expressed in the robot’s body frame at the current step, and the *direct* controller computes a proportional command

$$c_{\text{goal}} = (\text{clip}_{[-v_x^{\text{max}}, v_x^{\text{max}}]}(k_v x_{\text{local}}), \text{clip}_{[-v_y^{\text{max}}, v_y^{\text{max}}]}(k_v y_{\text{local}}), \text{clip}_{[-\omega^{\text{max}}, \omega^{\text{max}}]}(k_\psi \text{atan2}(y_{\text{local}}, x_{\text{local}}))), \quad (1)$$

which is the uncompensated baseline: a strong pretrained policy driven straight at the goal. Such proportional commands are the standard task layer on velocity-conditioned policies; they saturate far from the goal and induce energy-intensive in-place stepping near it. Compensation replaces c_{goal} when—and only when—a corrected command is trusted.

3.2 Command-Response Model

Adapters and audits both need predictions of what a candidate command would do. We model the *closed loop* of policy and robot at the command interface, not action-level dynamics [Nagabandi et al., 2018]: the policy already handles contacts, and re-modeling them would duplicate that work at far higher dimensionality. The closed-loop view keeps the input three-dimensional and makes identification a supervised problem on the policy’s own rollouts (R1, R3).

Excitation data. We override the converged policy with commands sampled throughout its admissible command space, including low-magnitude commands near the goal. This is persistent excitation applied to the policy–robot closed loop. Crucially, *no quality filtering is applied*: stumbles, slips, and low-posture episodes are kept, because the model must know where the policy performs poorly. Filtering to high-quality segments is a natural shortcut and an actively harmful one: a model trained only on clean behavior cannot warn the gate about the sloppy behavior it must avoid (Section 4.2).

Response targets. The outputs instantiate a general three-channel schema that any instantiation of the framework must fill: a *progress* statistic sufficient to score task advancement, the *cost* being minimized, and the *viability* quantity the gate guards. For locomotion, constant-command windows provide body-frame displacement $(\Delta x, \Delta y)$ and yaw change $\Delta\psi$ for progress; a mechanical cost for the objective; and minimum and mean terrain-relative posture height for viability. We use two cost channels of increasing physical fidelity. The first is a dimensionless *actuation-effort proxy* $\bar{a} = \text{mean}_{t,j}(|a_{t,j}\dot{q}_{t,j}|)$, formed from the public policy action a ; because a is not a measured motor torque, this proxy has no physical energy unit and is treated as an optimization objective, not an energy measurement. The second is a torque-derived mechanical power $\bar{P} = \text{mean}_{t,j}(|\tau_{t,j}\dot{q}_{t,j}|)$, where $\tau_{t,j}$ is the simulator-applied joint torque; this is a simulated mechanical quantity, still not battery energy, but the physically meaningful channel used to audit whether reducing the proxy reduces mechanical work. A manipulation instantiation would substitute end-effector displacement, an appropriate actuation cost, and a grasp-stability or joint-limit margin. Posture height is recovered from the policy’s own height-scan observation (mean scan return plus sensor offset), not from world coordinates. This prevents ground elevation from being confounded with posture; its empirical value is reported with the response-model results.

Model. The response model is a mapping

$$f_\phi : (o_t, c) \mapsto (\widehat{\Delta x}, \widehat{\Delta y}, \widehat{\Delta\psi}, \hat{C}, \hat{h}_{\min}, \hat{h}_{\text{mean}}), \quad (2)$$

where hats mark quantities *predicted* by the model, plain symbols are measured, and $\hat{C}$ is the chosen cost channel. The implementation takes the full policy observation with its command component replaced by the candidate command; architecture, training, and split details are given in Appendix A. Conditioning on the full observation is what makes the predictions *situational*: the same candidate command is predicted to cost differently on different terrain and from different gait states, which is precisely the information a fixed command-shaping heuristic cannot express.

3.3 Adapter Value Audit

Average response accuracy does not determine whether a learned adapter is worth deploying (Figure 1). The decision depends on three distinct quantities: whether alternative commands can help at the same state, whether their treatment effects are predictable from deployment-time information, and whether state-conditioned allocation beats issuing the same commands at the same frequencies without conditioning. We formalize these quantities before introducing any particular adapter.

Let x denote the observable query context (observation and task target), a_0 the direct command, and $\mathcal{A}$ a finite intervention set. Let $Y_x(a) = (W_x(a), T_x(a), S_x(a))$ be the potential mechanical work, completion time, and success after taking intervention a and following a common continuation. For a time budget ρ , candidate a is eligible when

$$E_x(a) = \mathbf{1}[S_x(a) \geq S_x(a_0), T_x(a) \leq \rho T_x(a_0)] = 1. \quad (3)$$

In simulation we observe these potential outcomes by exact start-state replay; without reset access they must instead be estimated from randomized or logged interventions, a limitation we return to in Section 5.

Available headroom. The best fixed action $a^g = \arg \min_{a \in \mathcal{A}} \mathbb{E}_x[W_x(a)]$ (subject to the declared aggregate constraints) measures global operating-point gain $H_{\text{global}} = 1 - \mathbb{E}W_x(a^g)/\mathbb{E}W_x(a_0)$. The same-state oracle $\pi^*(x) = \arg \min_{a: E_x(a)=1} W_x(a)$ instead measures the full local opportunity present in $\mathcal{A}$:

$$H_{\text{avail}} = 1 - \frac{\mathbb{E}_x[W_x(\pi^*(x))]}{\mathbb{E}_x[W_x(a_0)]}. \quad (4)$$

This is an oracle diagnostic, not a deployable controller.

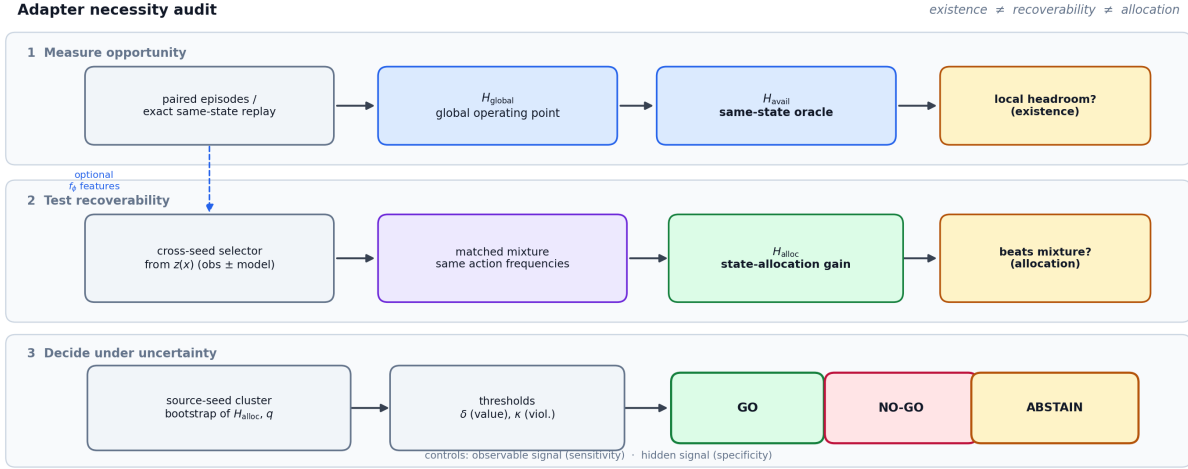


Figure 1: Adapter necessity audit. Stage 1 measures opportunity (H_{global} , H_{avail}). Stage 2 isolates state-allocation gain H_{alloc} over a frequency-matched mixture. Stage 3 maps source-seed uncertainty into GO/NO-GO/ABSTAIN under thresholds (δ , κ), with observable/hidden controls for sensitivity and specificity. Identified-model features f_ϕ are optional inputs to Stage 2.

Recovered and allocation value. A selector $\hat{\pi}(z(x))$ is cross-fitted across independent source seeds from deployment features z ; these may contain only observations or append identified response predictions. Its total recovered gain is $H_{\text{total}} = 1 - \mathbb{E}W_x(\hat{\pi}(x))/\mathbb{E}W_x(a_0)$. Total gain alone is insufficient because any selector can improve simply by issuing a globally cheaper scale more often. We therefore construct $\mu_{\hat{\pi}}$, which draws actions independently of x with exactly the marginal frequencies $p_a = \Pr_x[\hat{\pi}(x) = a]$ of the learned selector. The state-allocation value is

$$H_{\text{alloc}} = 1 - \frac{\mathbb{E}_x[W_x(\hat{\pi}(x))]}{\mathbb{E}_{x,a \sim p}[W_x(a)]}. \quad (5)$$

Only $H_{\text{alloc}} > 0$ is evidence that conditioning commands on context adds value beyond a matched non-adaptive mixture. We separately report the realized constraint-violation rate $q = \Pr_x[E_x(\hat{\pi}(x)) = 0]$.

Three-way deployment decision. Because a statistically positive but operationally negligible gain does not justify an adapter, the deployer specifies a minimum practical allocation value δ and maximum violation rate κ . We cluster-bootstrap whole source seeds, yielding confidence intervals $[L_H, U_H]$ for H_{alloc} and $[L_q, U_q]$ for q . The audit returns

$$\begin{aligned} \text{GO} &: L_H > \delta \wedge U_q \leq \kappa, \\ \text{NO-GO} &: U_H < \delta \vee L_q > \kappa, \\ \text{ABSTAIN} &: \text{otherwise.} \end{aligned} \quad (6)$$

The abstention region is essential: an underpowered audit is not evidence that an adapter is unnecessary. We report threshold sensitivity rather than selecting δ after observing the result.

Reusable protocol. Before data collection, an audit instantiation declares $(\mathcal{A}, W, E, \delta, \kappa)$ and the independent deployment unit to be resampled. It then (i) measures the fixed operating-point frontier, (ii) obtains paired potential outcomes on representative query contexts, (iii) cross-fits candidate selectors from deployment-available representations and evaluates each against its frequency-matched mixture, and (iv) applies Eq. 6 together with observable- and hidden-signal controls. This protocol is domain-independent; exact replay, mechanical work, and command scales are the locomotion instantiation used here.

3.4 Viability-Gated Compensation

VGCC scores a structured candidate set built around the direct command: uniform scalings of c_{goal} (factors 0.4 to 1.2), yaw-rate offsets, and a small polar grid of absolute speeds at heading offsets around the target direction, about 70 candidates in total, all clipped to the admissible command box (exact values in Appendix A). One batched forward

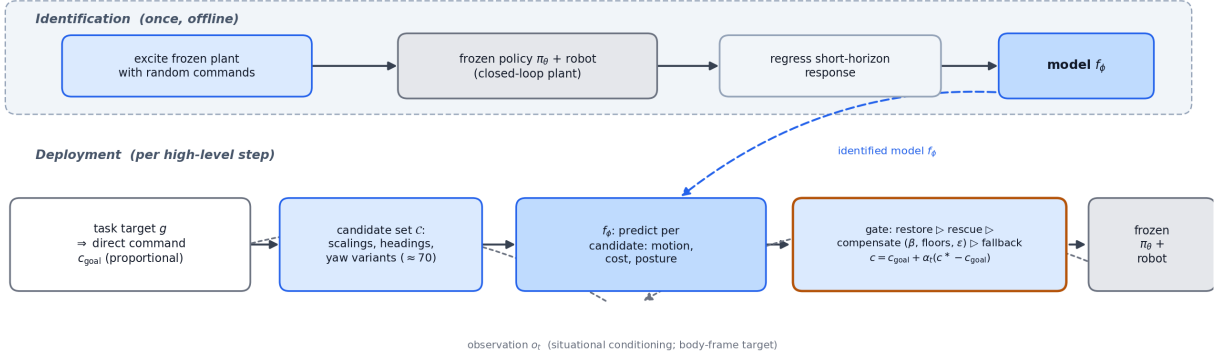


Figure 2: VGCC overview (case study under the audit). Offline excitation rollouts train an observation-conditioned response model. Online, the model scores a structured candidate set around the direct command; a priority-ordered gate applies a bounded correction only when progress, posture, and cost-margin checks pass, and otherwise uses exact direct fallback.

pass of f_ϕ predicts every candidate’s response. The set is wide enough to contain slower, redirected, and re-timed variants of the direct command, but each candidate is still a single command held for one decision interval, not a plan.

Scoring and execution. The gate scores raw candidates; what it executes is a bounded correction toward the selected candidate c^* , $c_{\text{exec}} = c_{\text{goal}} + \alpha_t (c^* - c_{\text{goal}})$, with gain $\alpha_t = \alpha \text{clip}(d/d_{\text{anneal}}, 0, 1)$ that anneals to zero as the remaining distance d shrinks, so corrections fade out on final approach. The executed command therefore always lies between the strong direct command and a model-certified candidate. Scoring the raw candidate (not the blend) is a simplification whose error is limited by the bounded gain.

Progress floor as a slowdown bound. Predicted target progress is

$$\hat{G}(c) = \|g\| - \|g - (\hat{\Delta}x(c), \hat{\Delta}y(c))\|. \quad (7)$$

A candidate is eligible when it preserves most of direct progress and posture:

$$\hat{G}(c) \geq \beta \hat{G}(c_d), \quad \hat{h}_{\min}(c) \geq h_{\text{floor}}, \quad \hat{h}_{\text{mean}}(c) \geq h_{\text{now}} - \delta, \quad (8)$$

with $c_d = c_{\text{goal}}$. Because the horizon has fixed duration, $\beta = 0.9$ caps instantaneous slowdown at roughly 10% and blocks the trivial “walk slower” solution; removing it does exactly that (Section 4.6).

Cost objective and margin. Among eligible candidates, VGCC selects $c^* = \arg \min_{c \text{ eligible}} \hat{C}(c)$ and intervenes only if

$$\hat{C}(c^*) < (1 - \epsilon) \hat{C}(c_d). \quad (9)$$

The progress floor bounds slowdown; ϵ rejects model noise. Normalizing cost by predicted progress performs worse in a paired ablation (Appendix B.3): with the floor already present, the extra normalization over-suppresses compensation.

Posture rescue and restore. Two reflexes complete the controller. If the model predicts that even the direct command will breach the posture floor within the horizon, VGCC instead selects, among candidates predicted to make non-negative progress, the one with the highest predicted minimum height (*rescue*); the bounded correction then pulls the executed command toward it. If measured posture has already sagged below the operating band, VGCC suspends compensation and issues the aggressive direct command until height recovers (*restore*). Both reflexes default toward, not away from, the strong direct controller, preserving R2. The model is thus a veto on candidate commands, not an unconstrained optimizer. All scalars are selected on development seeds and frozen before confirmation; Appendix A lists their values.

Pseudocode is given in Appendix A (Algorithm 1). The one-step gated comparison is intentional: f_ϕ is accurate for local near-stationary consequences (Section 4.2) but not for long-horizon compounding, and ϵ abstains when ranking sits at the edge of that accuracy. Section 4.7 audits whether average accuracy suffices for counterfactual ranking.

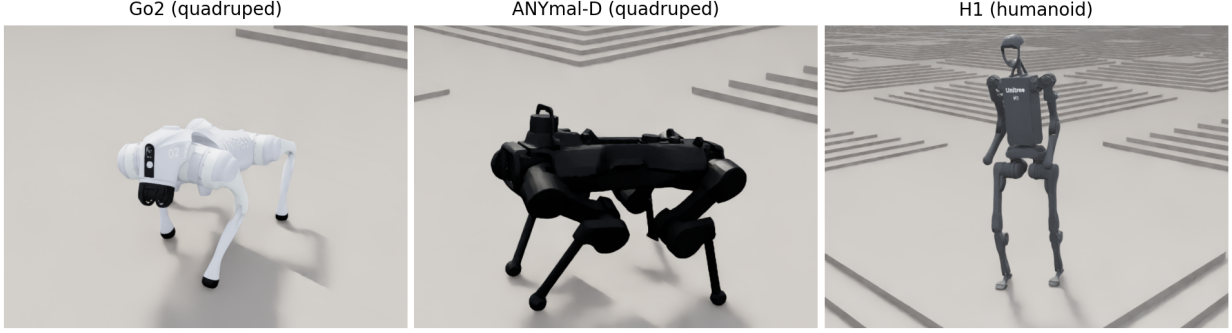


Figure 3: The three embodiments in their Isaac Lab rough-terrain velocity environments: the Go2 and ANYmal-D quadrupeds and the H1 humanoid. Each is driven by its publicly distributed pretrained RSL-RL policy, which is the policy VGCC compensates; VGCC changes only the command that policy receives, never its weights. The two quadrupeds share every gate parameter unchanged; the humanoid differs only in its posture thresholds.

4 Experiments

Experiments answer three questions in the order needed by the audit. **Q1:** how accurately can the closed-loop command response be identified (Section 4.2)? **Q2:** as a case-study adapter, how does VGCC compare with proportional control, tuned fixed scaling, a reactive governor, and sampling MPC (Sections 4.3–4.4, with coverage in Section 4.5)? **Q3 (central):** how much same-state headroom exists, how much is recoverable beyond a frequency-matched mixture, and what deployment decision does the audit return (Section 4.7)? Q1 establishes the optional feature source; Q2 places concrete adapters on the efficiency frontier; Q3 is the paper’s main claim—existence versus recoverability versus allocation.

4.1 Setup

Environments and robots. All experiments use Isaac Lab [Mittal et al., 2023] rough-terrain velocity environments with three embodiments (Figure 3): *Isaac-Velocity-Rough-Unitree-Go2-v0* and *Isaac-Velocity-Rough-Anymal-D-v0* (quadrupeds) and *Isaac-Velocity-Rough-H1-v0* (humanoid), with the publicly distributed pretrained RSL-RL checkpoints [Rudin et al., 2022, Schwarke et al., 2025] as the compensated policies. The task is target reaching: move the base to a target g specified relative to the start pose. Success requires no early termination, final XY distance below 0.3 m, and *final height* above 70% of spawn height. Final height is a terminal posture check, so that an episode that reaches the target crouched, collapsed, or mid-fall does not count as a success. Because the terrain is procedurally rough, final height reflects both posture and the ground elevation where the robot stops, which is why we report it alongside, rather than inside, the cost and distance metrics.

Metrics. We report success, final distance, final height, and how often compensation was active. The optimized running cost is the dimensionless actuation-effort proxy $\text{mean}_{t,j}(|a_{t,j}\dot{q}_{t,j}|)$; since the public policy action a is not measured motor torque, this proxy has no physical energy unit. To test whether reducing the proxy reduces mechanical work, a torque-derived evaluation records simulator-applied joint torques τ , integrates $\sum_{t,j} |\tau_{t,j}\dot{q}_{t,j}|\Delta t$, and reports mean mechanical power, work per traveled meter, cost of transport (COT, normalized by mg and path length), and elapsed task time (Sections 4.3 and 4.4).

Baselines. (1) *direct*: the pretrained policy under the direct target command, the policy being compensated and a strong baseline; (2) *fixed scaling*: the direct command multiplied by a constant scale (0.75 or 0.90), the open-loop efficiency heuristic; (3) *reactive governor*: a measured height/tilt governor that scales the command down when measured posture degrades. Random and zero-command baselines are excluded as uninformative against a strong pretrained policy. A retrieval-based compensator (VGSR) is reported in Appendix B.2.

Task families. *Harder* Go2 targets: (2, 0), (1.5, 1), (0, 1.25), (−0.75, 0), (1, −1), (0, −1.25), (−1, 0.75), (1.75, −0.75) m: long, diagonal, lateral, and backward goals covering the command space (the last three directions were added when the benchmark was expanded and were never used in any tuning). *Moderate* Go2 targets: (0.75, 0), (0.75, 0.75), (1.5, 0), (0, 0.75) m. *Holdout* Go2 directions (never used during any tuning): (1.25, 0.5), (0.5, 1), (−0.5, 0.5), (1.25, −0.5) m. ANYmal-D and H1 use the same eight harder targets as Go2.

Table 1: Go2 command-response model, held-out episode test metrics ($\sim 25k$ excitation windows, 16-step horizon). Terrain-relative height labels are essential for the viability gate.

Output	MAE	RMSE	R^2
Δx (m)	0.019	0.025	0.944
Δy (m)	0.019	0.025	0.942
$\Delta \psi$ (rad)	0.029	0.042	0.927
effort proxy (per step)	0.435	0.625	0.752
min. rel. height (m)	0.009	0.012	0.875
mean rel. height (m)	0.008	0.010	0.906
min. height, world-frame labels	0.039	0.092	0.247

Protocol. VGCC’s gate scalars (Section 3.4) were tuned on a *single* Go2 seed (301) on the harder set, then frozen for all Go2 and ANYmal-D evaluations. Excitation collection, identification, and every gate parameter are shared unchanged across the two quadrupeds. The humanoid retains the objective, cost margin, response-model architecture, and correction gain, but uses more conservative posture thresholds (eligibility, rescue, and restore fractions (0.93, 0.90, 0.96) and height-decline tolerance 0.8 cm, versus (0.85, 0.80, 0.90) and 2 cm for the quadrupeds), selected after diagnosing the humanoid failure mode in Section 4.5; H1 therefore tests morphology-adapted transfer, not zero-tuning transfer. All reported results use disjoint evaluation seeds: harder $5 \text{ seeds} \times 10 \text{ trials} \times 8 \text{ targets}$ (400 episodes per method), moderate 3×10 (120), holdout 3×5 (60), ANYmal-D $3 \times 5 \times 8$ (120), H1 $3 \times 5 \times 8$ (120). Per-robot response models are identified with the identical recipe (Appendix A). We report episode-level bootstrap 95% CIs as descriptive uncertainty summaries. Because episodes share a small number of environment seeds, these intervals do not establish equivalence or noninferiority; paired seed-level differences are reported separately, and success comparisons are phrased as “no observed aggregate loss” rather than as proof of equality.

4.2 Identifying the Command Response

Table 1 reports held-out episode test metrics for the Go2 response model. Displacement and yaw responses are accurately predicted ($R^2 \geq 0.92$); the effort proxy is predicted well enough to rank candidates ($R^2 = 0.75$); and terrain-relative posture height is predicted reliably ($R^2 = 0.88/0.91$ for min/mean, MAE under 9 mm), versus $R^2 = 0.25$ when the same model is trained on world-frame height labels, which terrain elevation corrupts. The excitation data matters as much as the architecture: a model trained on utility-filtered archive segments (the natural “reuse the archive” shortcut) reached $R^2 = -1.9$ on lateral displacement, because curated high-quality segments underrepresent exactly the sloppy behavior the gate must anticipate. The torque-derived mechanical-power channel used for the physical evaluations is identified with the same recipe by a five-member ensemble and reaches $R^2 = 0.797$ (MAE 13.0 W).

Identification data efficiency. Re-identifying the model from episode-group subsamples degrades gracefully: with 50% of the data, displacement R^2 holds at 0.93 and the effort channel drops to 0.72; with 25% ($\approx$ one minute of parallel simulation), displacement is still 0.92 and the effort channel 0.62. Displacement, which drives the progress constraint, saturates early; accuracy on the cost channel is what additional data improves. The 25% model is evaluated in closed loop in Appendix B.3.

4.3 Compensation on Go2

The harder-target benchmark comprises eight targets evaluated over $5 \text{ seeds} \times 10 \text{ trials}$, i.e. 400 episodes per method.

Running cost. VGCC reduces the actuation-effort proxy by 10.5% (1.976 vs. 2.208; descriptive episode-bootstrap CIs [1.938, 2.016] vs. [2.156, 2.256]). Observed success is similar (0.797, CI [0.760, 0.835], vs. 0.805, CI [0.765, 0.843]), as are final distance and height, with zero early terminations for either method. The experiment was not powered as an equivalence test, so CI overlap is not treated as evidence that success is statistically equal. The effect is directionally consistent *within* seeds: the paired per-seed difference is negative on all five benchmark seeds (-6.4% to -14.3%), and on all 17 evaluation seeds across the Go2 families, ANYmal-D, and H1 (two-sided exact sign test $p = 1.53 \times 10^{-5}$), while the paired success difference ranges from -0.05 to $+0.05$. Because the pooled sign test combines task families and embodiments, it supports consistency of direction, not a common effect size. Figure 4 visualizes both distributions.

Torque-derived mechanical work. To test whether the proxy reduction reflects mechanical effort, we re-evaluate the same configuration on three new seeds ($2 \text{ trials} \times 8 \text{ targets}$; 48 episodes per method) while recording simulator-applied joint torque (Table 2). Absolute mechanical work decreases on all three seeds (-8.4% , -9.4% , -6.6% ;

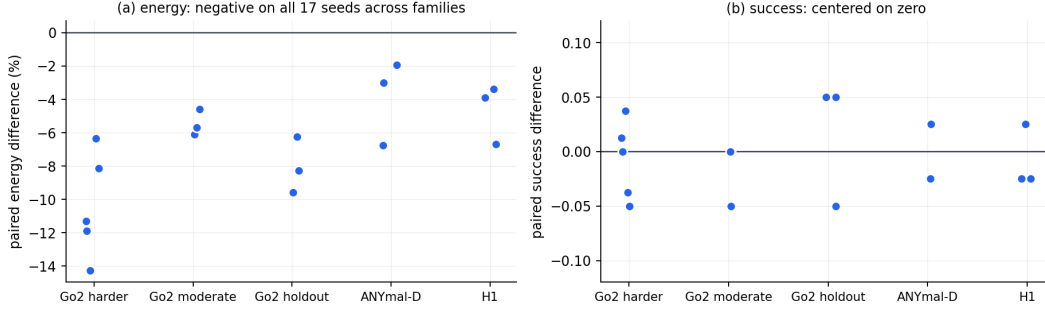


Figure 4: Paired per-seed differences (VGCC – direct) for all evaluation seeds. The effort-proxy difference is negative for 17/17 seeds (two-sided exact sign test $p = 1.53 \times 10^{-5}$). Success differences straddle zero. Points are descriptive seed aggregates; the pooled sign test combines heterogeneous families.

Table 2: Torque-derived mechanical-work evaluation of the main configuration on three new seeds ($n = 48$ episodes per method). Work is $\sum_{t,j} |\tau_{t,j} \dot{q}_{t,j}| \Delta t$ using simulator-applied torque; COT normalizes by mg and traveled path length. Values are descriptive because there are only three independent seeds.

Metric	VGCC	direct	Δ
Success	0.771	0.792	−0.021
Effort proxy	1.933	2.280	−15.2%
Absolute mechanical work (J)	129.89	141.48	−8.2%
Mean absolute power (W)	60.29	71.88	−16.1%
Elapsed task time (s)	2.113	1.905	+10.9%
Work per path length (J/m)	111.14	118.60	−6.3%
Cost of transport	0.754	0.805	−6.3%

−8.2% pooled). The decomposition is informative: VGCC lowers mean mechanical power by 16.1% but takes 10.9% longer, so the work saving is smaller than the power saving, and work per traveled meter and COT fall by 6.3%. Observed success differs by −2.1 points. This evaluation supports a simulated mechanical-work reduction for the main configuration; it is not an electrical or hardware energy claim, and its robustness to re-identification and to a tuned fixed scale is examined next.

4.4 Strong Baselines

A tuned fixed command scale is the simplest efficiency heuristic and the frontier any learned adapter must justify crossing. We compare against it, a reactive governor, and direct control on a locked six-seed evaluation with torque-derived power, all from identical initial states. VGCC is shown under both cost channels so the result does not depend on which channel is optimized (Table 3, Figure 5). Fixed 0.90 is the sharpest iso-time competitor; fixed 0.75 traces the slow, low-COT end.

VGCC reduces mechanical work (5.1%), power (16%), and COT (3.9%) below direct control, with the highest success of any method (0.802 vs. 0.792). It does *not* dominate the fixed-scaling frontier: at matched time, fixed 0.90 reaches lower COT (0.724 vs. 0.737), and fixed 0.75 the lowest COT (0.685) at a 38% time penalty. VGCC sits just above this frontier (Figure 5).

A heavier baseline does not close the gap. An 81-sequence sampling MPC plans four high-level steps over $\{0.75, 0.90, 1.0\}$ with a separately trained recursive macro-transition ensemble and task-value model (Appendix A). On three of the six seeds it reaches COT 0.757—below direct (0.765) but above VGCC (0.736) and fixed 0.90 (0.721). We therefore do not claim VGCC beats a well-tuned fixed scale. The remaining questions are transfer of the same configuration (Section 4.5) and whether state-conditioned allocation can justify adapter complexity over this frontier (Section 4.7).

Is the gain just a slowdown? COT and work per meter remove the trivial effect of traveling less distance, but are not generally speed-invariant. The progress floor is load-bearing: removing it deepens the proxy saving from 12.2% to 19.6% but costs 3.7 success points (Appendix B.3). VGCC lengthens episodes by 17% where fixed 0.75 pays 40%.

Table 3: Locked frontier comparison on six seeds ($n = 96/\text{method}$), ordered by completion time. All six methods are evaluated from *identical* initial states (paired resets), so the differences are within-condition. VGCC is shown under both cost channels. Bold marks the best value in each column. VGCC reduces work, power, and COT relative to direct control and attains the highest success, but a per-robot-tuned fixed scale traces a lower COT–time frontier: fixed 0.90 reaches a lower COT at matched time, and fixed 0.75 the lowest COT of all at a 38% time cost. A heavier recursive sampling MPC with separate transition and terminal-value models is discussed in the text.

Method	Success	Work (J) ↓	Power (W) ↓	Time (s) ↓	COT ↓
direct	0.792	135.07	68.09	1.918	0.767
fixed scaling (0.90)	0.781	126.85	57.28	2.145	0.724
VGCC (proxy cost)	0.802	128.20	56.99	2.207	0.737
VGCC (power cost)	0.802	128.28	56.67	2.220	0.737
fixed scaling (0.75)	0.781	119.42	44.05	2.648	0.685
reactive governor	0.771	128.51	44.69	2.947	0.740

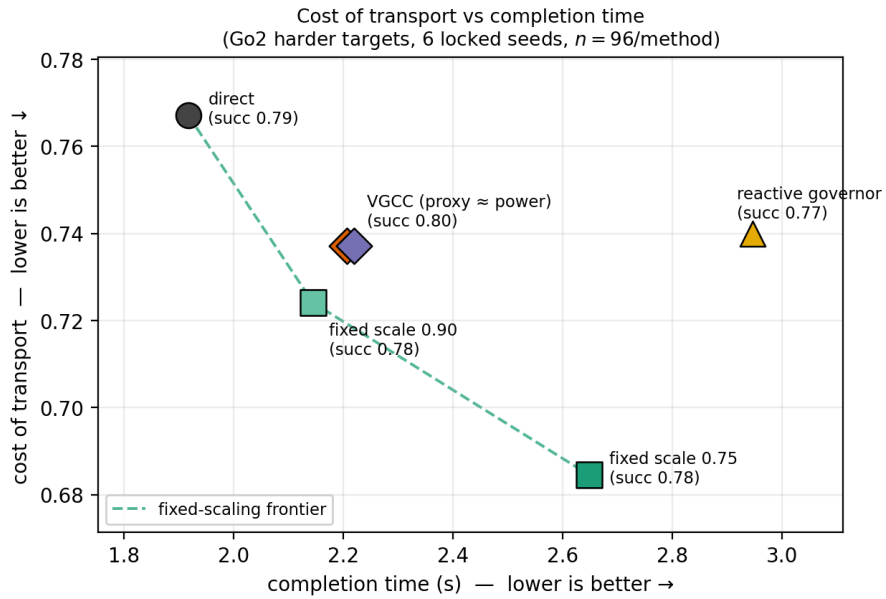


Figure 5: Cost of transport versus completion time on six locked seeds (Go2 harder targets, $n = 96/\text{method}$, identical initial states). The dashed line is the frontier traced by the direct policy and two fixed command scales. VGCC reduces COT relative to direct control but sits just above the fixed-scaling frontier, while attaining the highest success (0.80). The reactive governor is dominated. VGCC’s advantage is not frontier dominance but that a single configuration reaches this near-frontier operating point with no pre-selected scale, keeps the highest success, and (Section 4.5) transfers across the command space and across embodiments. Section 4.7 audits whether a learned state-conditioned allocation can justify its complexity over this frontier.

4.5 Coverage and Transfer

Before auditing recoverability, we check whether the VGCC case study is a Go2-harder artifact. Table 4 evaluates the frozen configuration on remaining Go2 target families and two further robots, without per-family or per-robot objective tuning.

Moderate and holdout targets. The effort proxy drops on both families (-5.4% and -8.0%), while observed success differs by -1.7 and $+1.6$ percentage points. The holdout family is the most informative case: these directions were never observed during any tuning, so the frozen gate generalizes across the command space, a property the retrieval variant does not possess (Appendix B.2: the archive variant *increases* the proxy on these directions).

Third embodiment, zero re-tuning: ANYmal-D. Applying the recipe unchanged to a different quadruped reduces the effort proxy by 3.9% ; observed success is 0.883 vs. 0.875 , with identical final posture and zero terminations. The paired effort difference is negative on all three ANYmal seeds. Since no gate or objective parameter was adjusted,

Table 4: Coverage across task families and embodiments. Δ is the relative change in the actuation-effort proxy. Success CIs are descriptive, not equivalence tests. Go2 and ANYmal-D use the frozen quadruped configuration; H1 uses morphology-adapted posture thresholds (Section 4.1).

Family (n /method)	Method	Success $\uparrow$	Effort $\downarrow$	Effort Δ	Final height (m)
Go2 moderate (120)	VGCC	0.975 [0.942, 1.000]	1.743 [1.689, 1.802]	−5.4%	0.356
	direct	0.992 [0.975, 1.000]	1.844 [1.763, 1.932]	N/A	0.362
Go2 holdout (60)	VGCC	0.933 [0.867, 0.983]	1.936 [1.836, 2.037]	−8.0%	0.334
	direct	0.917 [0.833, 0.983]	2.105 [1.982, 2.234]	N/A	0.333
ANYmal-D, 8 targets (120)	VGCC	0.883 [0.825, 0.942]	1.367 [1.317, 1.422]	−3.9%	0.500
	direct	0.875 [0.817, 0.933]	1.422 [1.373, 1.471]	N/A	0.500
H1, 8 targets (120)	VGCC	0.967 [0.933, 0.992]	0.690 [0.665, 0.716]	−4.7%	0.928
	direct	0.975 [0.942, 1.000]	0.724 [0.696, 0.752]	N/A	0.941

this is consistent with a cross-quadruped effect rather than a Go2-only artifact. A single robot and three seeds are nevertheless insufficient to establish broad embodiment generality.

H1 humanoid: posture floors are morphology-dependent. Transfer is not zero-shot. With quadruped posture thresholds unchanged, VGCC’s aggressive model-selected commands collapse H1 posture on long forward/diagonal targets (the 2 m target falls to 0.47): a taller COM and smaller support region cannot sustain the loose floors. Tightening eligibility, rescue, and restore thresholds—leaving the objective, cost margin, architecture, and correction gain unchanged—recovers a −4.7% effort-proxy reduction with success 0.967 vs. 0.975 and negative paired differences on all three seeds. Because thresholds were changed after observing the failure, this is morphology-adapted evidence, not zero-shot transfer.

4.6 Ablations

A paired component study (Appendix B.3, $n = 80$ per variant) isolates the role of each mechanism. Removing the posture gate forfeits most of the proxy saving (−4.8% vs. −12.2%) and reduces success; removing the progress floor deepens the saving to −19.6% but reduces success by 3.7 points; and full command substitution ($\alpha = 1$) has the largest success cost (−5.0 points), which is why the correction is bounded. The gate keeps predicted savings inside the model’s well-behaved posture regime, the progress floor limits behavioral drift, and bounded blending avoids the success cost of substitution. Quarter-data identification retains displacement $R^2 = 0.92$ but halves the realized proxy reduction, tracking the cost-channel degradation in Section 4.2. These results explain *how* VGCC achieves its direct-control improvement; they do not show that the improvement is state-allocation value. That distinction is the audit next.

4.7 Adapter Necessity Audit

Controller ranking alone cannot tell us whether an adapter failed because no state-dependent improvement exists or because its representation could not recover one. We therefore audit adapter necessity in three stages. Stage 1 measures *global versus episode-level* variation with paired full episodes. Stage 2 uses exact replay to measure *same-state* counterfactual headroom. Stage 3 cross-fits a treatment-effect selector and compares it with a randomized mixture that uses exactly the same commands at the same marginal frequencies. Only improvement over that mixture is evidence that conditioning on state adds value.

For every episode we have the ground-truth cost of transport of the direct command (1.0), fixed 0.90, and fixed 0.75 from identical initial states. We compare the best single *global* scale (minimizing pooled COT) against a finite-grid oracle that picks the best scale *per episode*. The oracle upper-bounds what any selector that holds one scale from this set for the whole episode could extract; the gap between it and the best global scale is the episode-level adaptivity headroom that exists to be won. A per-step switch (VGCC) or plan (the sampling MPC) is not formally bounded by an episode-level oracle, but neither undercuts even the best fixed scale in Table 3, let alone the oracle. Corrections outside the scale family are not searched by the oracle; VGCC’s candidate set does include redirected and re-timed variants, and their availability does not change its position against the fixed scales.

The episode-level gap is small (Figure 6): the per-episode oracle improves on the best single scale by only 1.0% on Go2 ($n = 72$ fully successful paired episodes; episode-bootstrap 95% CI [0.5, 1.7]%) and 1.9% on H1 ($n = 197$; CI [1.1, 3.0]%). H1 here is a second-embodiment Stage 1 screen with the same three-scale grid: the same near-constant

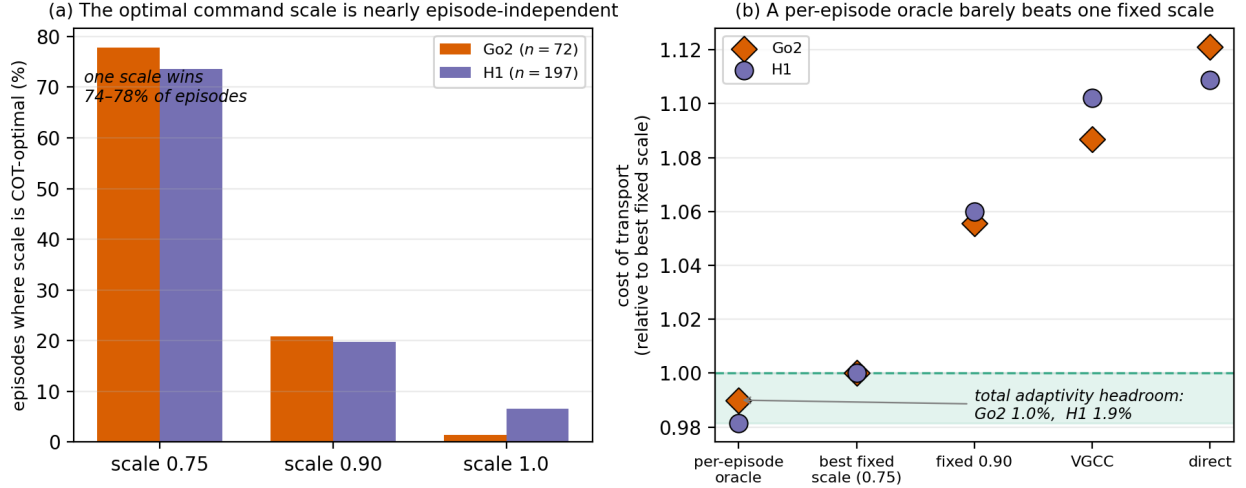


Figure 6: Stage 1 of the adapter necessity audit, on paired-successful episodes (identical initial states). **(a)** The COT-optimal command scale is nearly the same in every episode: a single scale is optimal in 74–78% of episodes on both robots. **(b)** Cost of transport relative to the best single fixed scale ($= 1$): a finite-grid per-episode oracle lies only 1.0% (Go2) and 1.9% (H1) below that fixed scale. This is an episode-level diagnostic over three scales, not an upper bound on within-episode switching or commands outside the grid.

Table 5: Adapter necessity audit on ten exact-replay source seeds (480 same-state queries, scales $\{0.75, 0.90, 1.00\}$, success no worse than direct, time $\leq 1.10 \times$ direct). The local oracle shows available headroom; the cross-fitted selector measures recoverable headroom; the matched mixture attributes the part due only to issuing slower commands more often.

Audit quantity	Work vs. direct	Constraint violations	Interpretation
same-state oracle	−5.21%	0% by construction	available local headroom
LOSO learned selector	−0.68%	2.08%	total selected gain
matched randomized mixture	−0.13%	3.93%	action-frequency gain
selector vs. matched mixture	−0.55%	—	state-allocation gain

global slowdown pattern. Stages 2–3 exact-replay confirmation on H1 is reported below. The per-episode winners show why: the most aggressive available slowdown wins 78% of Go2 episodes and 74% of H1 episodes, and COT falls monotonically as the scale drops, so the efficiency-optimal command is a near-constant global slowdown, not an episode-dependent choice. This result alone does *not* bound within-episode switching, redirected commands, or a richer command set. It is a cheap first screen: if a deployment only permits one scale per episode, learning a selector over this grid is unlikely to repay its complexity.

Stages 2–3: does local headroom exist, and is it recoverable? We next isolate a single intervention from long-horizon controller effects. At 480 query states from ten source seeds, each candidate prefix $s \in \{0.75, 0.90, 1.00\}$ is replayed from the *same* observation and then followed by the same direct-control continuation. Replay is exact after freezing the terrain curriculum: the query-observation reconstruction error is zero. A candidate is eligible only if its realized success is no worse than direct and its remaining completion time is at most 1.10 times direct. The same-state oracle then chooses the eligible prefix with minimum total mechanical work. Unlike the episode-level analysis, this test shows meaningful local variation: the oracle reduces work by 5.21% while slightly increasing success (0.698 versus 0.681).

Existence is not predictability. We train a five-member treatment-effect ensemble to predict each candidate’s work and time *relative to direct*, using the deployment observation, target, and identified macro-response means and uncertainties. Every reported prediction is leave-one-source-seed-out. A conservative selector intervenes only when the work upper bound is negative, the time upper bound satisfies the budget, and the eligibility lower bound exceeds 0.5. To separate state information from action frequency, its matched randomized mixture draws scales independently of state using the selector’s own scale frequencies within each held-out fold.

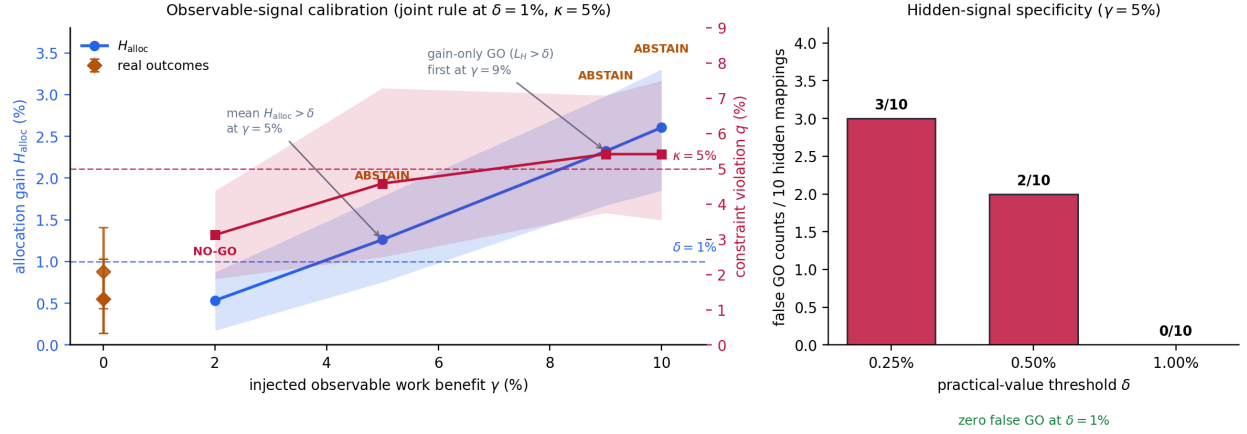


Figure 7: Calibration of the three-way decision on ten source seeds. **Left:** observable-signal sweep. Mean H_{alloc} exceeds $\delta = 1\%$ by $\gamma = 5\%$, and the gain-only rule ($L_H > \delta$, ignoring q) first returns GO at $\gamma = 9\%$; the joint rule with $\kappa = 5\%$ abstains throughout because stronger bonuses inflate the violation CI. Real outcomes ($\gamma = 0$) abstain. **Right:** ten independent 5% hidden-signal mappings; false GO counts fall to zero at $\delta = 1\%$.

The learned selector activates on 13.3% of queries and matches the oracle choice on 46.3%. Its candidate relative-work MAE is 11.6%, versus only 5.2% total oracle headroom and 0.55% recovered state-allocation gain. It beats the matched mixture on work in seven of ten held-out seeds, but the effect remains below one percent. With source-seed cluster bootstrap, $H_{\text{alloc}} = 0.55\%$ has a descriptive 95% CI of $[0.14, 1.03]\%$, and $q = 2.08\%$ has CI $[1.04, 3.33]\%$. For a deployer requiring $\delta = 1\%$ allocation value and allowing $\kappa = 5\%$ violations, Eq. 6 returns ABSTAIN at every tested $\delta \in \{0.25\%, 0.5\%, 1\%\}$: the upper bound now reaches just above 1%, so a five-seed NO-GO does not survive doubling the independent units. An observation-only selector on the same ten seeds is firmer at $\delta = 1\%$ ($H_{\text{alloc}} = 0.41\%$ with CI $[0.16, 0.68]\%$; NO-GO), and adding the recursive macro predictions raises the point estimate only from 0.41% to 0.55%. The five-seed feature ablation in Appendix A.1 reaches the same qualitative conclusion. Identification is therefore tested as an optional source of prospective decision features, rather than credited with a bound that a ground-truth sweep computes without it.

Second embodiment (H1 Stages 2–3). We repeat the exact-replay protocol on the H1 humanoid with four source seeds (192 queries; replay observation error zero). Collection uses the same constant-scale prefixes; episode driving uses dimension-matched stub macro and task-value networks so that MPC plumbing can run, while the audit itself is *observation-only* and does not depend on stub predictions (Appendix A). At $\rho = 1.10$ the same-state oracle reduces work by 4.18% (success 0.927 vs. 0.906). A leave-one-seed-out observation selector recovers only 0.61% versus direct and 0.33% beyond its matched mixture (H_{alloc} CI $[0.04, 0.66]\%$; $q = 1.04\%$ with CI $[0.0, 2.08]\%$; candidate work MAE 6.7%). At $\delta = 1\%$ and $\kappa = 5\%$ the audit returns NO-GO—firmer than Go2’s ten-seed ABSTAIN, and consistent with Go2’s observation-only NO-GO. Thus the existence–recoverability gap is not a Go2 idiosyncrasy.

Can the audit ever return GO? The Go2 real benchmark is a low-value case, so we fill the sensitivity/specificity gap with semi-synthetic controls that keep all 480 real query states, outcomes, and cross-seed splits (Figure 7; Table 6). These calibrate the decision rule; H1 above is the second *robot* domain. For the *observable-signal* control, scale 0.90 receives a fractional work bonus on nonnegative-lateral targets and scale 0.75 otherwise; the pre-intervention target exposes which scale is preferred. We sweep the bonus γ from 2% to 10%. A *hidden-signal* control assigns the preferred scale per query and omits it from all features, creating oracle headroom no selector should systematically recover.

Mean H_{alloc} exceeds $\delta = 1\%$ once $\gamma = 5\%$, and a gain-only rule ($L_H > \delta$, ignoring violations) first returns GO at $\gamma = 9\%$ ($L_H = 1.67\%$). The joint rule never returns GO in the tested 2%–10% range: stronger bonuses raise intervention rates and push U_q above κ . The five-seed calibration that returned GO at 9% therefore does not survive the larger cluster sample—a feature of the declared decision, not a detection failure. Ten independent 5% hidden-signal mappings produce three false GO decisions at $\delta = 0.25\%$, two at 0.5%, and zero at 1% (four NO-GO, six ABSTAIN). Thus 1% is the smallest tested threshold with zero false GO in this calibration, not a universal constant. Full curves are in Appendix A.2.

Table 6: Calibration of the three-way decision (ten source seeds, 100,000 source-seed bootstrap replicates, $\delta = 1\%$, $\kappa = 5\%$). Observable-signal rows inject a known context-dependent work benefit into otherwise unchanged replay outcomes.

Condition	H_{alloc} , 95% CI	q , 95% CI	Decision
real outcomes	0.55 [0.14, 1.03]%	2.08 [1.04, 3.33]%	ABSTAIN
observable signal, 2%	0.53 [0.17, 0.87]%	3.12 [1.88, 4.38]%	NO-GO
observable signal, 5%	1.26 [0.75, 1.78]%	4.58 [2.50, 7.29]%	ABSTAIN
observable signal, 9%	2.32 [1.67, 2.98]%	5.42 [3.75, 7.08]%	ABSTAIN
observable signal, 10%	2.61 [1.85, 3.30]%	5.42 [3.54, 7.50]%	ABSTAIN

Does terrain heterogeneity open the frontier? A natural conjecture is that mixed terrain would: if rough ground rewards slowing more than flat ground, a terrain-conditioned scale should beat any single fixed one. We test this with flat and uniformly rough (2–10 cm) variants, a re-identified response model, and the scale grid extended to 0.60 (Appendix A, Table 7). COT falls monotonically with the scale on *both* terrains: roughness raises the cost level but barely moves where its minimum lies, so no winner flips. A shared completion-time budget that forces choosing *where* to slow down yields only 1.6% of total work (bootstrap 95% CI [0.8, 8.4]) at the tightest budgets, and nothing once slowing is allowed everywhere. The conjectured mechanism does not open the frontier on the roughness axis.

We are careful about scope. Episode-level figures are grid-specific. Exact-replay Stages 2–3 cover Go2 (ten seeds) and H1 (four seeds, observation-only); H1 collection uses stub closed-loop models only to drive episodes, not as audit features. Terrains we did not vary (slopes, compliance, friction) could reshape rather than merely shift the cost curve; richer interfaces such as gait parameters [Margolis and Agrawal, 2022] are where tracking may not flatten structure. The claim is not that adaptivity can never help, but that its headroom is measurable—and on this benchmark family, recoverable allocation gain is small.

5 Limitations and Future Work

Energy validity. Large-sample results optimize a dimensionless actuation-effort proxy; torque audits support lower simulated mechanical work and COT versus direct control, but not versus the fixed-scaling frontier. Absolute joint work is not electrical energy; hardware power measurement remains necessary for a real energy claim.

Time entanglement. VGCC episodes run 10–14% longer despite the progress floor. An explicit completion-time budget would make residual savings unambiguously iso-time; the mixed-terrain shared-budget estimate is only 1.6% of work with a wide CI [0.8, 8.4].

Statistics. Episode bootstrap intervals ignore nesting in seeds; overlapping success CIs are not equivalence tests. The Go2 adapter audit resamples ten source seeds; the H1 confirmation uses four. Thresholds and seed counts should be pre-registered for confirmatory studies.

Recoverability, not impossibility. Local oracle headroom of 5.2% (Go2) / 4.2% (H1) with recovered allocation gains of 0.55% / 0.33% is evidence about the present representation and data, not a theorem that no learner can recover the oracle. Semi-synthetic controls (Figure 7) fill decision-rule sensitivity and specificity, including a gain-only GO at $\gamma = 9\%$. H1 Stages 2–3 provide a second-robot confirmation at $\delta = 1\%$ (NO-GO). Remaining gaps: hardware energy, and H1 audits that include trained identified-model features rather than observation-only.

Scope. All results are simulated target reaching with velocity-commanded policies on three embodiments. The framework assumes a command interface with redundancy and fast closed-loop response. Untested: end-effector-commanded manipulation, style-commanded character control, hardware transfer, and richer gait interfaces where the tracking objective may not flatten state-dependent structure.

6 Conclusion

Before building a learned command adapter on a frozen locomotion policy, one should measure whether the interface supports it. We contribute an adapter necessity audit that separates global operating-point gain, same-state counterfactual headroom, and recoverable state-allocation gain over a frequency-matched mixture, then returns GO/NO-GO/ABSTAIN under practical-value and violation thresholds. Closed-loop identification provides optional features and is accurate on average, but average accuracy does not imply treatment-effect ranking: on Go2, 5.2% local work headroom yields only 0.55% allocation gain; on H1, 4.2% yields 0.33% with a $\delta = 1\%$ NO-GO. That gap—not empty

local opportunity—explains why VGCC and sampling MPC sit above a tuned fixed-scaling frontier. With ten Go2 seeds the real audit abstains at $\delta = 1\%$; semi-synthetic controls (Figure 7) show a gain-only GO at $\gamma = 9\%$ and zero false GO at $\delta = 1\%$ under hidden signal, while the joint rule abstains once violations become uncertain. Next steps: hardware energy, and H1 audits with trained identified-model features.

Reproducibility

All experiments use public Isaac Lab environments and public pretrained RSL-RL checkpoints. Excitation collection, response-model training, evaluation, and plotting code, together with all result files aggregated in this paper, are available in the project repository.

References

- Tom Silver, Kelsey Allen, Josh Tenenbaum, and Leslie Kaelbling. Residual policy learning. *arXiv preprint arXiv:1812.06298*, 2018.
- Tobias Johannink, Shikhar Bahl, Ashvin Nair, Jianlan Luo, Avinash Kumar, Matthias Loskyll, Juan Aparicio Ojea, Eugen Solowjow, and Sergey Levine. Residual reinforcement learning for robot control. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 6023–6029, 2019.
- Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. In *Robotics: Science and Systems (RSS)*, 2021. arXiv:2107.04034.
- Gabriel B. Margolis and Pulkrit Agrawal. Walk these ways: Tuning robot control for generalization with multiplicity of behavior. In *Conference on Robot Learning (CoRL)*, 2022. arXiv:2212.03238.
- Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Conference on Robot Learning (CoRL)*, pages 91–100, 2022. arXiv:2109.11978.
- Viktor Makoviychuk, Lukasz Wawrzyniak, Yunrong Guo, Michelle Lu, Kier Storey, Miles Macklin, David Hoeller, Nikita Rudin, Arthur Allshire, Ankur Handa, and Gavriel State. Isaac Gym: High performance GPU based physics simulation for robot learning. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2108.10470.
- Mayank Mittal, Calvin Yu, Qinxi Yu, Jingzhou Liu, Nikita Rudin, David Hoeller, Jia Lin Yuan, Ritvik Singh, Yunrong Guo, Hammad Mazhar, Ajay Mandlekar, Buck Babich, Gavriel State, Marco Hutter, and Animesh Garg. Orbit: A unified simulation framework for interactive robot learning environments. *IEEE Robotics and Automation Letters*, 8(6):3740–3747, 2023. arXiv:2301.04195.
- Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning quadrupedal locomotion over challenging terrain. *Science Robotics*, 5(47):eabc5986, 2020.
- Takahiro Miki, Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science Robotics*, 7(62):eabk2822, 2022.
- Antonio Sciarretta, Giovanni De Nunzio, and Luis Leon Ojeda. Optimal ecodriving control: Energy-efficient driving of road vehicles as an optimal control problem. *IEEE Control Systems Magazine*, 35(5):71–90, 2015.
- Emanuele Garone, Stefano Di Cairano, and Ilya Kolmanovsky. Reference and command governors for systems with constraints: A survey on theory and applications. *Automatica*, 75:306–328, 2017.
- Marc Peter Deisenroth and Carl Edward Rasmussen. PILCO: A model-based and data-efficient approach to policy search. In *International Conference on Machine Learning (ICML)*, pages 465–472, 2011.
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018. arXiv:1805.12114.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019. arXiv:1906.08253.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:1912.01603.
- Nicklas Hansen, Xiaolong Wang, and Hao Su. Temporal difference learning for model predictive control. In *International Conference on Machine Learning (ICML)*, 2022. arXiv:2203.04955.
- Anusha Nagabandi, Gregory Kahn, Ronald S. Fearing, and Sergey Levine. Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 7559–7566, 2018. arXiv:1708.02596.

- Kim P. Wabersich and Melanie N. Zeilinger. A predictive safety filter for learning-based control of constrained non-linear dynamical systems. *Automatica*, 129:109597, 2021.
- Tairan He, Chong Zhang, Wenli Xiao, Guanqi He, Changliu Liu, and Guanya Shi. Agile but safe: Learning collision-free high-speed legged locomotion. In *Robotics: Science and Systems (RSS)*, 2024. arXiv:2401.17583.
- Jeonghwan Kim, Yunhai Han, Harish Ravichandar, and Sehoon Ha. Learning Koopman dynamics for safe legged locomotion with reinforcement learning-based controller. *arXiv preprint arXiv:2409.14736*, 2024.
- Karl Johan Åström and Richard M. Murray. *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press, 2008.
- Bruce A. Francis and W. Murray Wonham. The internal model principle of control theory. *Automatica*, 12(5):457–465, 1976.
- David Q. Mayne, James B. Rawlings, Christopher V. Rao, and Pierre O. M. Scokaert. Constrained model predictive control: Stability and optimality. *Automatica*, 36(6):789–814, 2000.
- Grady Williams, Paul Drews, Brian Goldfain, James M. Rehg, and Evangelos A. Theodorou. Information-theoretic model predictive control: Theory and applications to autonomous driving. *IEEE Transactions on Robotics*, 34(6):1603–1622, 2018.
- Juraj Kabzan, Lukas Hewing, Alexander Liniger, and Melanie N. Zeilinger. Learning-based model predictive control for autonomous racing. *IEEE Robotics and Automation Letters*, 4(4):3363–3370, 2019.
- Jared Di Carlo, Patrick M. Wensing, Benjamin Katz, Gerardo Bledt, and Sangbae Kim. Dynamic locomotion in the MIT Cheetah 3 through convex model-predictive control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1–9, 2018.
- Toru Kitagawa and Aleksey Tetenov. Who should be treated? empirical welfare maximization methods for treatment choice. *Econometrica*, 86(2):591–616, 2018.
- Susan Athey and Stefan Wager. Policy learning with observational data. *Econometrica*, 89(1):133–161, 2021.
- Adith Swaminathan and Thorsten Joachims. Counterfactual risk minimization: Learning from logged bandit feedback. In *International Conference on Machine Learning*, pages 814–823, 2015.
- Richard S. Sutton, Doina Precup, and Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1-2):181–211, 1999.
- Pierre-Luc Bacon, Jean Harb, and Doina Precup. The option-critic architecture. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, 2017.
- Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. Diversity is all you need: Learning skills without a reward function. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1802.06070.
- Archit Sharma, Shixiang Gu, Sergey Levine, Vikash Kumar, and Karol Hausman. Dynamics-aware unsupervised discovery of skills. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:1907.01657.
- Xue Bin Peng, Yunrong Guo, Lina Halper, Sergey Levine, and Sanja Fidler. ASE: Large-scale reusable adversarial skill embeddings for physically simulated characters. In *ACM Transactions on Graphics (SIGGRAPH)*, volume 41, 2022. arXiv:2205.01906.
- Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.
- Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023. arXiv:2304.13705.
- Clemens Schwarke, Mayank Mittal, Nikita Rudin, David Hoeller, and Marco Hutter. RSL-RL: A learning library for robotics research. *arXiv preprint arXiv:2509.10771*, 2025.

Algorithm 1 Viability-Gated Command Compensation (one high-level step)

Require: frozen policy π_θ , response model f_ϕ , target g , correction gain α , anneal distance d_{anneal} , and fixed scalars $(\beta, h_{\text{floor}}, h_{\text{resc}}, \delta, \epsilon)$

- 1: Compute body-frame target g , direct command c_{goal} (Eq. 1), and the candidate set $\mathcal{C}$ around c_{goal}
- 2: Batch-predict $f_\phi(o, c)$ for all $c \in \mathcal{C}$; set $\alpha_t = \alpha \text{clip}(d/d_{\text{anneal}}, 0, 1)$
- 3: **if** measured posture is below the operating band **then**
- 4: $c \leftarrow c_{\text{goal}}$ $\triangleright$ restore
- 5: **else if** direct control is predicted to breach the posture floor h_{resc} **then**
- 6: $c^* \leftarrow \arg \max_{\mathcal{C}: \hat{C} \geq 0} \hat{h}_{\min}$; $c \leftarrow c_{\text{goal}} + \alpha_t(c^* - c_{\text{goal}})$ $\triangleright$ rescue
- 7: **else**
- 8: $c^* \leftarrow \arg \min_{\mathcal{C} \text{ eligible (Eq. 8)}} \hat{C}$
- 9: **if** $\hat{C}(c^*) < (1 - \epsilon) \hat{C}(c_{\text{goal}})$ (Eq. 9) **then**
- 10: $c \leftarrow c_{\text{goal}} + \alpha_t(c^* - c_{\text{goal}})$ $\triangleright$ compensate
- 11: **else**
- 12: $c \leftarrow c_{\text{goal}}$ $\triangleright$ exact direct fallback
- 13: **end if**
- 14: **end if**
- 15: Execute π_θ with c for E low-level steps

A Implementation Details

All values below are the ones used for the paper’s experiments; the exact launch commands are in the project repository.

Excitation and dataset. Excitation overrides the converged policy’s command with commands drawn uniformly over the admissible box $[-1, 1]^3$ in (v_x, v_y, ω_z) , resampled every 25 low-level steps; with probability 0.3 the drawn command is scaled by 0.3, oversampling the low-magnitude regime that proportional control visits near the goal. Collection runs 64 parallel environments (8,000 steps per environment for H1). Constant-command windows of 16 low-level steps (0.32 s at the 50 Hz control rate) are sliced at stride 4 within episodes, giving the $\approx 25\text{k}$ Go2 windows of Table 1. Terrain-relative height labels are recovered from the policy’s own height-scan observation (mean scan return plus its 0.5 m sensor offset).

Response model and training. f_ϕ is a multilayer perceptron with two hidden layers of width 256 (ReLU, dropout 0.1), mapping the full policy observation, with the command slice replaced by the candidate command, to the six response targets; inputs and targets are standardized. Training uses AdamW (learning rate 10^{-3} , weight decay 10^{-4}), batch size 1,024, 200 epochs, keeping the checkpoint with the best held-out loss. The held-out set is 20% of episode groups, so no episode straddles the split; Table 1 reports this held-out set. The torque-derived mechanical-power channel is identified with the same recipe by a five-member ensemble.

Candidate set. The structured set $\mathcal{C}$ around the direct command comprises: uniform scalings $\{0.4, 0.6, 0.8, 1.2\} c_{\text{goal}}$ and c_{goal} itself; yaw-rate offsets of $\{\pm 0.2, \pm 0.4\}$ rad/s applied to c_{goal} ; and a polar grid of absolute speeds $\{0.25, 0.5, 0.75, 1.0\}$ m/s at heading offsets $\{0, \pm 0.3, \pm 0.6\}$ rad around the target bearing with yaw-rate scalings $\{0, 0.5, 1.0\}$ of the direct yaw command. All candidates are clipped to the admissible box $[-1, 1]^3$ and deduplicated, leaving ≈ 70 .

Controller constants. One high-level decision is taken every $E = 4$ low-level steps. The gate scalars shared by all three robots are: correction gain $\alpha = 0.5$ with anneal distance $d_{\text{anneal}} = 0.5$ m (the correction fades out linearly inside 0.5 m of the target), progress floor $\beta = 0.9$, and cost margin $\epsilon = 0.1$; the cost channel $\hat{C}$ is the effort proxy for the main configuration and predicted mechanical power for the power configuration of Table 3. The morphology-dependent posture floors are the eligibility, rescue, and restore fractions of nominal height with the height-decline tolerance δ : (0.85, 0.80, 0.90) with $\delta = 2$ cm for both quadrupeds, (0.93, 0.90, 0.96) with $\delta = 0.8$ cm for H1 (Section 4.1).

VGCC pseudocode. Algorithm 1 states one high-level decision. If no candidate is eligible, or the cost margin fails, the controller executes c_{goal} exactly.

Sampling MPC baseline. The MPC of Section 4.4 does *not* recursively use the six-output response model f_ϕ : that model has no next-observation output and cannot be rolled forward. Instead, MPC uses a separately trained five-member macro-transition ensemble

$$F_\eta(o_t, c_t) \mapsto (\widehat{o_{t+4}} - o_t, \widehat{\Delta x}, \widehat{\Delta y}, \widehat{\Delta \psi}, \widehat{P}, \widehat{h}_{\min}, \widehat{\theta}_{\max}, \widehat{\omega}_{\max}),$$

where one macro step is four low-level steps (0.08 s). The model has two hidden layers of width 512 and is trained on 42,054 windows from 207 episode groups with disjoint train/validation/test groups. Each ensemble member recursively advances its own predicted full observation residual; the command slice is overwritten by the next planned command before the following query. At four macro steps (0.32 s), held-out open-loop errors are 3.3 cm displacement MAE, 5.17 J cumulative-work MAE, 8.8 mm minimum-height MAE, and 0.756 normalized full-state RMSE.

MPC enumerates all $3^4 = 81$ scale sequences from $\{0.75, 0.90, 1.00\}$ and executes the first command of the selected sequence. After the four-step rollout, a second, separately trained task-value network predicts remaining work, remaining time, and success under direct-control continuation from the predicted terminal observation and local target. It is trained on 4,298 samples from 144 mixed-controller episodes; held-out metrics are work MAE 15.7 J ($R^2 = 0.846$), time MAE 0.362 s ($R^2 = 0.654$), and success AUC 0.819. MPC minimizes ensemble-upper-bound total work subject to predicted success, time, posture, and optional stability constraints, with exact direct-sequence fallback.

Thus MPC shares the decision rate and command bounds with VGCC but not its model or data budget. Its 81 sequence evaluations are numerically close to VGCC’s approximately 70 one-step candidates, but each sequence uses four recursive ensemble transitions plus a terminal-value query. We describe it as a heavier model-based baseline, not a same-model or equal-compute comparison.

Mixed-terrain evaluation (Section 4.7). The flat and rough single-type tasks replace the terrain generator of the otherwise unchanged rough environment with a single sub-terrain each: a plane, and uniform height noise of 2–10 cm; every scene, observation, command, and event setting is identical, and the same pretrained checkpoint is used. The response model for these deployments is identified once on pooled excitation from both terrains with the unchanged recipe ($\approx 25k$ windows; displacement $R^2 = 0.92$, cost 0.73). Fixed scales $\{0.60, 0.75, 0.90, 1.00\}$ and VGCC run from identical initial states on three seeds disjoint from all tuning. Because the world-frame final-height criterion is dominated by stop-point ground elevation on ± 10 cm noise, completion means reaching the target without early termination. Table 7 reports the paired frontier: COT falls monotonically on both terrains. The budget analysis pairs one flat with one rough leg, sweeps a shared completion-time budget over the feasible range, and compares the best budget-feasible uniform scale against the best per-terrain scale allocation on the convex hulls of the per-terrain (work, time) means; the reported interval is an episode bootstrap over both legs.

Table 7: Single-terrain-type paired evaluation (Go2, three seeds, eight targets $\times$ two trials; completion = target reached without early termination; flat $n = 48$, rough $n = 36$ jointly completed episodes per scale, of 48 evaluated). COT falls monotonically with the scale on both terrains.

Scale	Flat			Rough (2–10 cm)		
	COT	Work (J)	Time (s)	COT	Work (J)	Time (s)
1.00	1.027	179.6	1.80	1.670	290.5	2.16
0.90	0.964	166.7	1.99	1.598	278.5	2.30
0.75	0.897	152.6	2.36	1.513	265.2	2.76
0.60	0.882	148.8	2.90	1.447	247.8	3.31

Exact-replay adapter necessity audit. The Go2 audit of Section 4.7 uses ten source seeds, 48 query states per seed (480 total), and constant four-macro-step prefixes from $\{0.75, 0.90, 1.00\}$. For each query and scale, the simulator is reset to the episode start, the recorded direct-control prefix is replayed to the query, the candidate prefix is executed, and direct control completes the episode. The terrain curriculum is frozen; all query observations reconstruct exactly (maximum observation error zero). This start-state replay is necessary because Isaac Lab’s scene-state reset does not restore policy observation and action history. Seeds 854, 855, 856, 858, and 859 are the original collection; seeds 860, 862, 863, 864, and 865 were added with the same protocol (`run_exact_replay_prefix_seeds.sh`). The exploratory four-way feature ablation table below remains on the original five-seed subset; the main text reports the ten-seed observation-only versus observation+ F_η comparison.

H1 second-embodiment exact replay. The H1 confirmation uses the same constant-scale prefix protocol (`run_exact_replay_prefix_seeds_h1.sh`) with `-ff_height_scan_slice 69:256`. Because trained H1 macro-transition and task-value models were not available, collection drives episodes with dimension-matched untrained stubs (`h1_macro_transition_h4_stub5.pt`, `h1_task_value_stub.pt`). These stubs affect only which query states are visited; every reported counterfactual outcome is an exact same-state replay with observation reconstruction error zero. The H1 treatment-effect audit uses the observation-only feature set with `-height-scan-slice 69:256` and does not consume stub prefix predictions. Four completed source seeds (870–873; 192 queries) are reported in the main text; additional seeds continue under the same recipe.

The same-state oracle requires candidate success no worse than the direct branch and total remaining time no greater than 1.10 times direct, then minimizes total mechanical work. The treatment-effect model is a five-member, two-layer MLP ensemble (hidden width 64) trained for 200 epochs. Inputs contain the deployment observation, local target, candidate scale, and differences between candidate and direct macro-response ensemble predictions (work, progress, minimum height, maximum tilt, and maximum angular speed; means and uncertainties). Targets are candidate work and time relative to the direct branch and realized eligibility. Evaluation leaves one entire source seed out. The selector accepts a candidate only when its work upper confidence estimate is negative, its time upper estimate meets the budget, and its eligibility probability lower estimate is at least 0.5.

For each held-out fold, the matched randomized mixture uses the selector’s exact marginal frequencies of scales 0.75, 0.90, and 1.00, but assigns them independently of query state. Its reported metrics are the exact expectation over the replayed potential outcomes, so no Monte Carlo sampling noise is introduced. The selector–mixture difference isolates state-dependent allocation from the gain due merely to issuing slower commands more often. Scripts and aggregate artifacts are `analyze_exact_replay_oracle.py`, `train_exact_replay_treatment_effect.py`, `exact_replay_prefix_oracle_5seed.json`, and `exact_replay_adapter_necessity_audit_5seed.json`.

A.1 Treatment-Effect Feature Ablation

We isolate which deployment-time representation contributes to the final decision while holding the treatment-effect ensemble, five leave-one-source-seed-out folds, confidence rule, and matched-mixture evaluation fixed. All variants receive the current observation summary, local target, and candidate scale. The f_ϕ variant adds candidate-versus-direct means and ensemble uncertainties from the six-output one-window response model. The F_η variant adds the corresponding four-step macro-prefix predictions used by MPC.

Table 8: Exploratory feature ablation for the exact-replay treatment-effect selector (240 queries). “Allocation gain” is selector work relative to its own frequency-matched randomized mixture; negative is better. The pre-existing main result is `observation+ $F_\eta$` .

Features	Work MAE	Active	Work vs. direct	Allocation gain	Violations
observation only	12.60%	8.8%	−0.43%	−0.38%	0.42%
observation + f_ϕ	12.93%	13.8%	−0.37%	−0.35%	4.17%
observation + F_η	12.22%	12.9%	−0.54%	−0.42%	1.67%
observation + f_ϕ + F_η	12.33%	12.1%	−0.76%	−0.76%	2.92%

Observation-only features recover nearly the same allocation gain as the prespecified `observation+ $F_\eta$` representation (0.38% versus 0.42%). Adding f_ϕ alone does not improve work error or allocation and increases realized constraint violations. Combining both identified models gives the largest point estimate, 0.76%, but this interaction was discovered on the same five folds, is below one percent, and incurs more violations; it is not a nested or independently confirmed model-selection result. We therefore retain `observation+ $F_\eta$` as the main prespecified audit and report the combined row only as a hypothesis for future confirmation. The robust conclusion is that the treatment-effect ensemble extracts most of its limited signal from the observation itself; neither identified model provides a decisive incremental contribution on this dataset.

A.2 Audit Calibration Controls

The semi-synthetic calibration changes only candidate mechanical-work outcomes; observations, targets, success, time, posture, source-seed splits, and the treatment-effect learner remain unchanged. For observable controls, the preferred scale is 0.90 when the pre-intervention local target has $y \geq 0$ and 0.75 otherwise, and its realized work is multiplied by $1 - \gamma$. Hidden controls draw the preferred scale independently for each query using a deterministic seed and do not expose that draw in any feature. The matched-mixture calculation is repeated after each modification.

Confidence intervals resample the ten source-seed folds as clusters with 100,000 bootstrap replicates (seed 20260717). Table 9 reports the ten-seed observable-signal sweep used in the main text. Decisions use $\kappa = 5\%$; G, N, and A denote GO, NO-GO, and ABSTAIN.

Relative to the earlier five-seed curve, the work intervals move as expected once $\gamma \geq 5\%$, but the violation intervals are wider and cross $\kappa = 5\%$, so the joint rule abstains throughout this sweep. A five-seed GO at $\gamma = 9\%$ therefore does not survive the larger cluster sample: underpowered audits can manufacture apparent detection.

To estimate specificity, we repeat the 5% hidden-signal control with ten independent mappings (control seeds 20260720–20260729) on the same ten source seeds. The false-GO counts are 3/10, 2/10, and 0/10 at $\delta =$

Table 9: Observable-signal calibration on ten source seeds. Decision columns correspond to practical allocation-value thresholds δ .

γ	$H_{\text{alloc}}, 95\% \text{ CI}$	Mean q	$\delta = .25\%$	.50%	1.00%
2%	0.53 [0.17, 0.87]%	3.12%	A	A	N
5%	1.26 [0.75, 1.78]%	4.58%	A	A	A
9%	2.32 [1.67, 2.98]%	5.42%	A	A	A
10%	2.61 [1.85, 3.30]%	5.42%	A	A	A

0.25%, 0.50%, 1.00%, respectively. At $\delta = 1\%$, four mappings return NO-GO and six ABSTAIN. This stress test motivates reporting the entire decision curve and keeping δ no smaller than the threshold with zero false GO outcomes in calibration. It does not establish a universal 1% threshold: another domain requires its own deployment value and control-based calibration.

The control experiments are generated by `train_exact_replay_treatment_effect.py`; source-seed intervals and decisions are produced by `summarize_adapter_value_audit.py`. Aggregate artifacts are `adapter_value_audit_observable_curve_10seed_summary.json` and `adapter_value_audit_hidden_05pct_repeats_10seed_summary.json`. The prefix collection recipe is `run_exact_replay_prefix_seeds.sh`.

B Secondary Experiments and Diagnostics

This appendix retains material that is useful for diagnosis and reproducibility but is not required for the main claim: qualitative and per-target results, a retrieval-based alternative, and the complete component ablation.

B.1 Qualitative and Per-Target Results

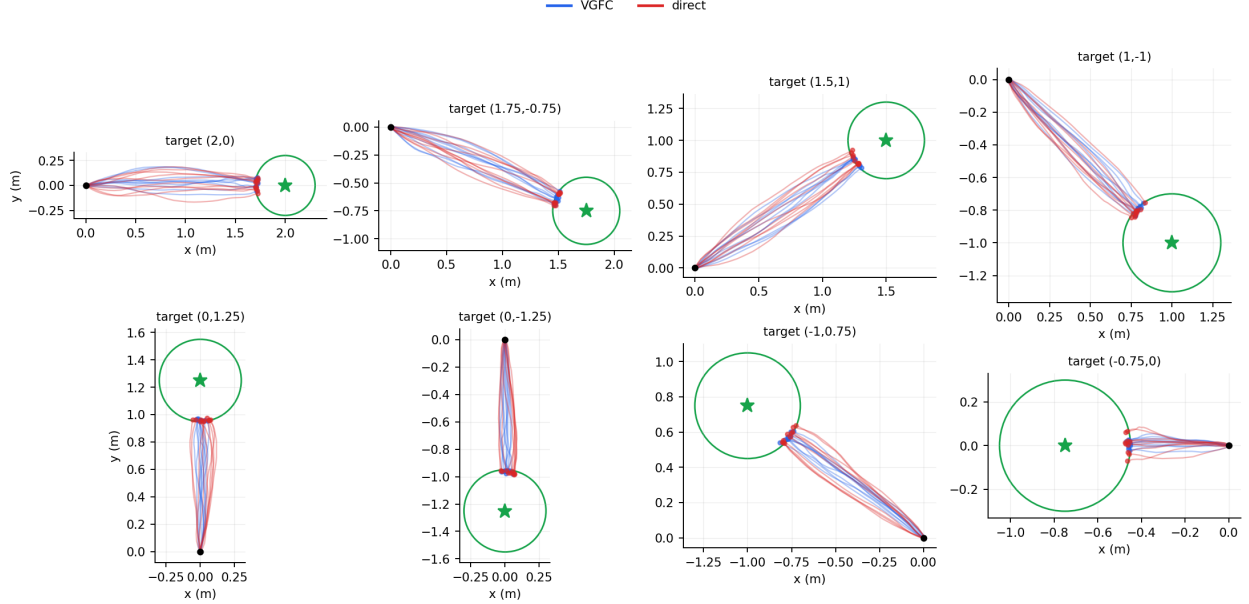


Figure 8: XY paths on the eight harder targets (one evaluation seed, 10 trials per method). VGCC and direct control reach similar target neighborhoods.

The selected example forecasts a posture-floor breach and activates rescue, but the aggregate residual failure mode is low-posture arrival. A symmetric posture-aware stopping rule did not improve success at $n = 400$ per method; it instead spent the remaining budget attempting recovery. Individual examples must therefore not be interpreted as a success-rate effect; the aggregate evidence is in Tables 2 and 4.

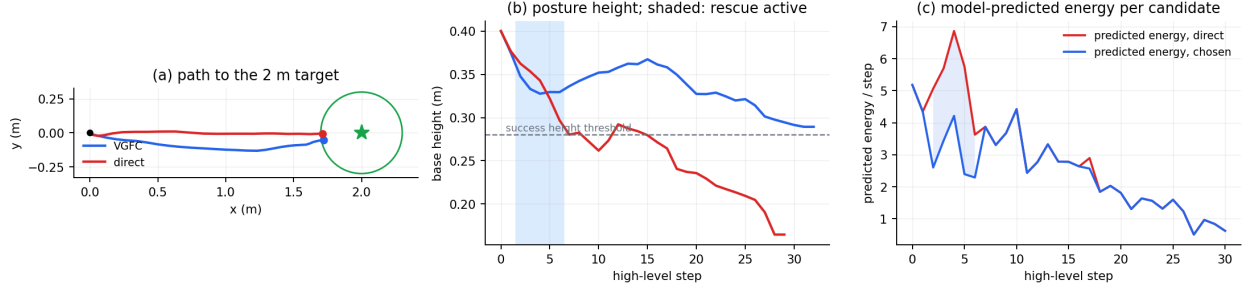


Figure 9: Selected paired episode illustrating rescue behavior. This example shows mechanism, not prevalence.

Table 10 gives the per-target breakdown of the harder benchmark. The proxy reduction holds on seven of eight targets, but success differences change sign at $n = 50$ per target, which is why we report aggregate success rather than a target-specific success effect.

Table 10: Per-target harder-target results (50 episodes per target and method), ordered by proxy reduction.

Target	Effort proxy ↓			Success ↑		Final height (m)	
	VGCC	direct	Δ	VGCC	direct	VGCC	direct
(0, 1.25)	1.815	2.083	−12.9%	0.940	0.980	0.340	0.351
(1.75, −0.75)	2.217	2.542	−12.8%	0.480	0.520	0.276	0.285
(−1, 0.75)	1.861	2.122	−12.3%	1.000	1.000	0.358	0.367
(0, −1.25)	1.822	2.073	−12.1%	0.940	0.980	0.345	0.350
(1.5, 1)	2.203	2.471	−10.8%	0.800	0.740	0.316	0.313
(2, 0)	2.367	2.619	−9.6%	0.240	0.260	0.233	0.224
(1, −1)	2.028	2.240	−9.5%	0.980	0.960	0.355	0.349
(−0.75, 0)	1.496	1.510	−0.9%	1.000	1.000	0.359	0.363

B.2 Retrieval Alternative (VGSR)

An earlier, nonparametric alternative to the response model retrieves reusable behavior chunks instead of predicting command responses. VGSR segments converged-policy rollouts into chunks, describes them by body-frame displacement, velocity, yaw, posture, effort proxy, stability, and smoothness, then novelty-filters and clusters them into a behavior archive. At deployment it retrieves a chunk by progress, state applicability, effort, stability, and utility, and blends the chunk velocity with the direct command only when a heuristic gate accepts it. Unlike VGCC, it can interpolate only inside archive support.

On the original five-target protocol, VGSR matches VGCC’s proxy level on the harder family (2.061 versus 2.073) but has 1.2% terminations and lower final height. It *increases* the proxy by 3.8% on held-out directions and 4.4% on H1, exactly the generalization the parametric model provides and the archive does not. In a separate five-seed harder-target aggregate it reduces the proxy by 7.8% with success 0.785 versus 0.800. Removing its gate increases the proxy by more than 50%. Archive size is non-monotonic: the top 2,000 chunks improve success relative to all 3,904, while top-500 minimizes proxy at lower success. These coverage and curation dependencies motivate the parametric response model used by VGCC.

B.3 Complete VGCC Component Ablation

The gate keeps predicted savings inside the model’s well-behaved posture regime; the progress floor limits behavioral drift; and bounded blending avoids the larger success cost of substitution. A quarter of the identification data retains displacement $R^2 = 0.92$ but reduces the cost-channel R^2 from 0.75 to 0.62, halving the realized proxy reduction. An absolute proxy margin fails across embodiments (it increases the H1 proxy by 13.5%), so the relative margin in Eq. 9 is retained.

Objective ablation. A variant that ranks candidates by predicted cost *per predicted progress* (with execution-aligned scoring, all other mechanisms unchanged) was evaluated paired against the standard gate on the locked frontier proto-

Table 11: Paired component ablations on two seeds ($n = 80$ per variant). Differences are against direct control in the same runs.

Variant	Δ proxy $\downarrow$	Δ success	Active steps
VGCC (full)	-12.2%	+0.012	11.7
no annealing	-12.2%	+0.013	12.3
no viability gate	-4.8%	-0.025	8.9
no progress floor	-19.6%	-0.037	19.8
substitution ($\alpha = 1$)	-14.0%	-0.050	16.3
25% identification data	-6.4%	-0.012	10.7

col (three seeds, $n = 48$ episodes per method, identical initial states). It forfeits most of the compensation: effort proxy 2.010 versus 1.810 for the raw-cost gate (direct 2.132), mechanical work 132.3 versus 126.9 J, success 0.792 versus 0.854, and it intervenes less (elapsed time 2.08 versus 2.25 s). The per-seed paired proxy difference favors the raw-cost gate on all three seeds. With the progress floor already capping slowdown, dividing by predicted progress double-penalizes slower candidates, so the variant rejects interventions the floor would have certified; the anti-slowdown role belongs to the constraint, not the objective.